Productive Ignorance in AI-Assisted Decision-Making

A Socratic Teammate, an Inspectable Architecture, and a Protocol for Comparing Advice and Inquiry

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Abstract

AI decision support can make uncertain situations appear settled, encouraging people to accept an answer before examining its assumptions. We investigate *productive ignorance* as a design principle for supporting judgment: recognising missing evidence and unresolved objections, and using them to guide inquiry rather than force closure. Our goal is to determine whether assistance can compensate for limitations in an individual's weaker decision environment while preserving effective behaviour in another. We present a prototype that makes task evidence and uncertain participant interpretations inspectable, retains objections through defeasible argumentation, and separately decides whether an intervention warrants interrupting the person. The design combines a no-assistance baseline with a 2×2 comparison of direct advice versus Socratic inquiry and generic versus personalised assistance, spanning multi-armed bandit (MAB) and target-based sequential (TBS) tasks. An implemented empirical pipeline imports unassisted histories for 62 individuals, preserves a 16-person development subset and a 46-person evaluation complement, reproduces the original fitted MAB model and constructs a provisional TBS policy. Historical TBS offers are unavailable and remain unknown. Preliminary held-out prediction improves on a choice-frequency baseline for MAB on average, but not for TBS; this limits claims of behavioural fidelity. A development pilot holds response strength fixed while varying three Socratic processing mechanisms, exposing how conclusions depend on assumed reconsideration. Its full-protocol results remain pending. The programme provides a controlled basis for examining when AI should advise, question or remain silent to support contestability and human judgment in national security and governance, without establishing human assistance effects.

Draft status. Software architecture, empirical modelling procedure, reconsideration-sensitivity rationale and completed engineering checks as of 6 November 2026. The full-protocol development pilot is in progress; its aggregate outcomes and human assistance effects are not reported. Prepared as a basis for a future contribution to the Stanford discussion on *The Future of Decision–Making: Examining the human implications of decision-making technologies in national security and governance.*

1 Motivation: supporting judgment without manufacturing closure

The practical question for AI decision support extends beyond the accuracy of its recommendations. A system can influence which uncertainties a person notices, which alternatives receive attention and whether a decision appears settled. In national security and governance, these effects concern responsibility and institutional accountability as well as task performance. Stanford's Future of Decision-Making project explicitly foregrounds human judgment, responsibility, over-reliance and meaningful human control in such settings [1]. We take this orientation as motivation for

studying the interaction mechanism, without treating a laboratory task as a model of an entire institution.

Current large language models (LLMs) characteristically produce fluent, coherent explanations, including when the available evidence is incomplete or supports competing interpretations. In decision support, this fluency can make a provisional interpretation appear settled: a person may accept a recommendation without recognising its dependence on missing evidence, fragile assumptions or unresolved disagreement. We call this risk *premature epistemic closure*. Retrieval and citation can improve access to evidence, but do not by themselves establish whether that evidence addresses the decision, whether competing interpretations remain viable, or whether an intervention is useful at that moment. The design challenge is therefore to make recognised uncertainty consequential for the system's behaviour. We adopt *productive ignorance* as a principle for retaining these limits and using them to guide inquiry, qualify advice or withhold an intervention, so that support helps the person examine the grounds for a decision before treating it as settled.

We operationalise productive ignorance through three commitments:

- First, observations, model estimates, hypotheses, attributed reports and unknowns remain distinguishable.
- Second, objections can defeat or leave an interpretation undecided rather than being absorbed into a fluent explanation.
- Third, the system must justify its opportunity to intervene: uncertainty alone does not establish that a question will be useful. Silence is a recorded policy outcome, not a missing message.

This position does not assume that questions outperform recommendations. Work on cognitive forcing functions illustrates that the form of an intervention can affect over-reliance [10]; it does not establish the effectiveness of our questions or their transfer to governance. A question can distract, suggest that a sound strategy is defective, or indirectly steer a choice. Direct advice can be useful when its criterion and limitations are explicit. Our comparison is designed to admit benefit, no effect and interference for either style.

1.1 Research questions and intended contribution

We ask two substantive questions:

1. Can assistance compensate for limitations in an individual's weaker decision environment while preserving effective behaviour in another environment?
2. How do assistance style and personalisation affect this outcome, for whom, under which task conditions and at what interruption cost?

We investigate these questions in two complementary decision-making contexts grounded in established experimental research. *Multi-armed bandit (MAB)* tasks are a standard paradigm for studying the exploration–exploitation trade-off: people must balance choosing options that have yielded rewards against sampling alternatives whose value remains uncertain [2]. *Target-based sequential decision-making (TBS)* situates choices within a sequence in which accumulated resources, a target and the remaining opportunities jointly affect which risks are worth taking. It belongs to a research tradition examining how people adapt risk-taking to these contextual demands [3]. Together, the tasks distinguish learning what the available options offer from adjusting a strategy as the constraints on action evolve. These demands connect to how decision-support technologies shape human judgment, responsibility and meaningful control in national security and governance. The motivating parallels include deciding whether further

evidence gathering warrants delaying action, revising a strategy when feedback challenges earlier expectations, and allocating limited resources towards an objective as opportunities diminish. MAB and TBS let us examine whether AI advice or inquiry helps maintain attention to these considerations, or instead encourages premature closure around a favoured option or an inappropriate decision criterion.

The present contribution is an executable architecture, an empirical actor-construction pipeline and a validation protocol for investigating these questions. Unassisted data from previous studies now constrain individual task policies, while the models' fidelity and responses to assistance remain separate questions. The pipeline supports repeated, controlled comparisons of five conditions that connect the proposed forms of support to the problem of premature epistemic closure:

- **A: No assistance.** Participants decide from the task information and their own experience, without AI advice or questions. This baseline allows us to examine whether assistance improves decisions or introduces interference through its framing and interruptions.
- **C: Generic direct advice.** The system uses current task information and observed history to recommend a specific option, without calibrated participant estimates. Advice states its comparison criterion and evidential limits, allowing us to examine whether a recommendation can support action while keeping its provisional grounds visible.
- **D: Generic Socratic inquiry.** The system uses task information and observed history to ask about relevant evidence, alternatives or constraints, without calibrated participant estimates or a prescribed option. Productive ignorance takes the form of a focused invitation to examine what remains unresolved before settling on a choice.
- **E: Personalised direct advice.** The system also uses uncertain estimates from the individual's unassisted calibration to select and time recommendations. This explores whether advice can address task-specific decision patterns while retaining uncertainty about the person, rather than turning a modelled tendency into a settled diagnosis.
- **F: Personalised Socratic inquiry.** The system combines task evidence with uncertain participant estimates to select and time questions about potentially overlooked evidence or assumptions. This explores whether targeted inquiry can help address an individual's limitations while leaving both the decision and the system's interpretation open to reconsideration.

Figure 1 summarises this design. All four assisted conditions retain the distinction between observations and interpretations, preserve relevant objections and permit silence when support is insufficient or interruption is unwarranted. Productive ignorance is therefore a shared design commitment whose expression varies across the conditions; the comparison does not presume that asking questions will prevent premature closure or outperform qualified advice.

Our aim is to make extensive use of the existing empirical evidence to refine the prototype's participant estimates, reasoning rules and intervention policies before conducting a new human assistance study. Repeated simulations with the same reconstructed individuals and fresh outcome realisations will allow us to examine how these design choices interact with different decision-making profiles and task demands. The previous unassisted data constrain the models of task behaviour; responses to assistance remain explicit assumptions to be varied and subsequently tested with people. A methodological contribution is the separation of the synthetic participant, its assumed response to assistance, the assistant's uncertain participant model, graph reasoning and delivery policy, making each component available for inspection and refinement.

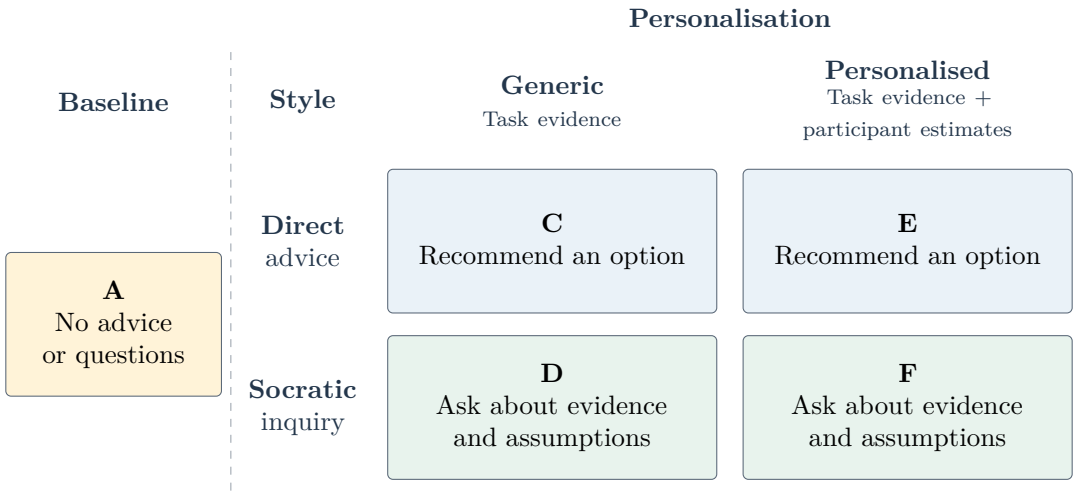


Figure 1: Five-condition comparison: A provides the no-assistance baseline outside the 2×2 matrix, which crosses communication style with personalisation. Generic assistance remains task-aware; personalised assistance additionally uses uncertain participant estimates. All assisted conditions retain uncertainty and objections and may remain silent.

2 Empirical setting and task fidelity

Our MAB context uses changing reward contingencies, making the interpretation of feedback and the revision of learned expectations central to performance. The full protocol consists of 200 choices among six options, with 50 initially stable trials and 150 volatile trials. Outcome probabilities are hidden. Participants infer values from observed feedback and balance exploration against exploitation. Current prototype options offer gain/zero/loss outcomes at magnitudes of 100 or 200 points; a missed choice incurs a 200-point penalty. Persistence after disappointing feedback can reflect noisy outcomes, a deliberate strategy or a failure to adapt. The observation alone does not distinguish these interpretations.

In our TBS context, two alternatives display their possible gains and win probabilities, but the consequences of a choice depend on progress towards a block target and the contracting horizon. The full protocol consists of 192 choices in 24 eight-trial blocks. A final-trial win multiplies the accumulated block total; a final-trial loss does not activate that multiplier. A missed response consumes an opportunity without a direct point penalty. Immediate expected reward and probability of attaining the target are distinct criteria.

The original tasks use a five-second response limit. The prototype also provides short engineering protocols of 40 MAB trials and 24 TBS trials. These shortened runs support testing and familiarisation; they cannot be substituted for the original protocols when claiming empirical reconstruction fidelity.

2.1 Empirical cohort and multilevel configurations

The collaboration draws on an existing experimental programme using two decision environments [5, 6]. The research materials describe a 62-person cohort with continuous, task-specific profiles combining psychological dispositions, resting-state and task-evoked neural measures, computational measures and behavioural policies. Task performance was examined separately and was excluded from profile scoring and assignment. Four exploratory multilevel configurations describe how the MAB-related and TBS-related profiles are jointly expressed:

- **MAB-specialised.** More pronounced expression of the MAB-related profile, with lower

expression of the TBS-related profile. This configuration emphasises the combination of measures associated with decision-making under uncertain and changing reward contingencies.

- **TBS-specialised.** More pronounced expression of the TBS-related profile, with lower expression of the MAB-related profile. This configuration emphasises the combination of measures associated with adjusting decisions to targets, accumulated resources and remaining opportunities.
- **Dual-regime.** Pronounced expression of both task-related profiles. The configuration captures the coexistence of the distinct multilevel patterns associated with the two decision environments, without implying that the same mechanisms support both.
- **Low-expression.** Lower expression of both task-related profiles. This describes the position of an individual on the constructed profile dimensions; it does not by itself establish generally poor decision-making or an absence of relevant abilities.

Continuous task-specific scores accompany these descriptions. The configurations are descriptive rather than validated fixed personality types, and neither their neural components nor SFSS dispositional scores establish responsiveness to AI. The supplied handover [7] resolves the current registry at 62 individuals with complete trial sequences for both tasks; earlier integrated analyses using 60 are not silently treated as the same analysis sample. We retain the original core, L1–L5 classifications and scores, excluding the performance layer from profile assignment. Their cohort counts are 21 MAB-specialised, 24 TBS-specialised, 10 Dual-regime and seven Low-expression. These labels describe the source cohort; they do not prescribe an actor's task policy or response to assistance.

3 Experimental design: five conditions and staged validation

3.1 Factorial comparison and baseline

The implemented condition set is shown in Table 1. Generic means *task-aware but non-personalised*. The earlier task-unaware guidance condition has been removed and is not part of this design. Condition letters are retained for continuity with the software.

Condition	Communication	Permitted task evidence	Participant calibration
A	No assistance	Available to the task runner	Withheld from assistance
C	Direct recommendation	Current public state and observed history	Withheld
E	Direct recommendation	Same task evidence	Used with uncertainty
D	Socratic question	Current public state and observed history	Withheld
F	Socratic question	Same task evidence	Used with uncertainty

Table 1: A no-assistance baseline plus a 2×2 comparison of style and personalisation. All conditions complete the required unassisted calibration; only E/F use its participant estimates to select assistance.

For a prespecified outcome Y, oriented so that larger values mean improvement, the factorial contrasts within each task are

$$\Delta_{\text{style}} = \tfrac{1}{2}[(Y_D - Y_C) + (Y_F - Y_E)], \tag{1}$$

$$\Delta_{\text{person}} = \tfrac{1}{2}[(Y_E - Y_C) + (Y_F - Y_D)], \tag{2}$$

$$\Delta_{\text{interaction}} = (Y_F - Y_E) - (Y_D - Y_C). \tag{3}$$

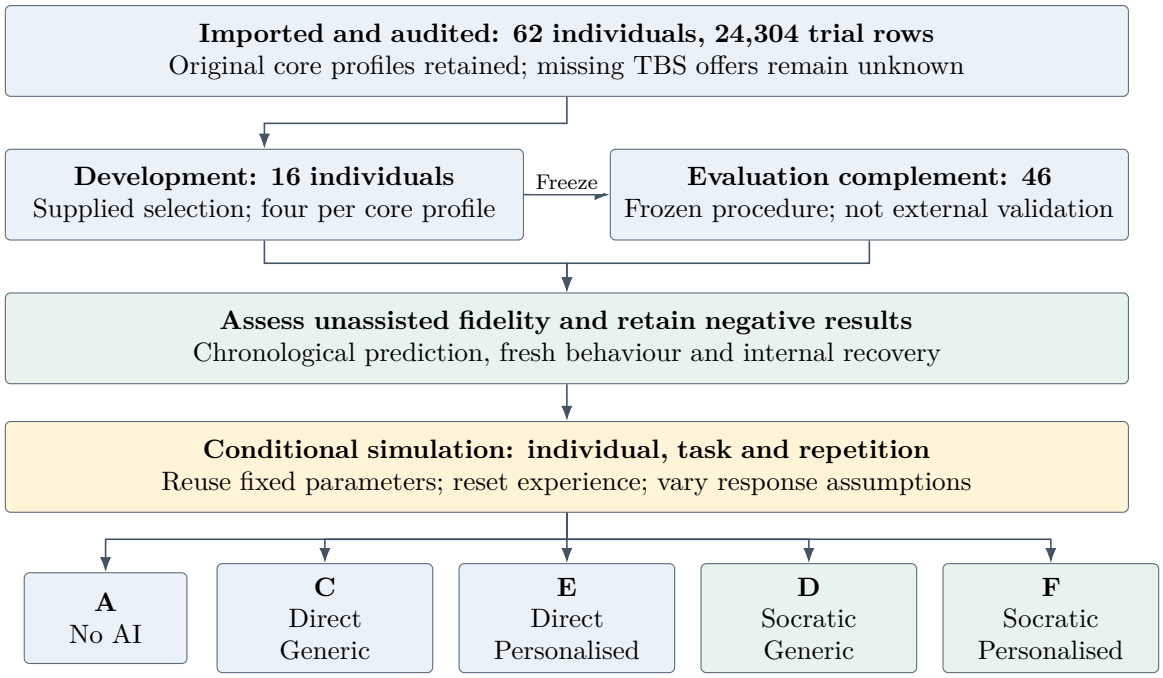


Figure 2: Implemented cohort pipeline and matched simulation design. Fidelity checks do not automatically validate an actor: the current TBS policy fails to improve on a simple frequency baseline on average. Assistance runs remain conditional on model and response assumptions. The reset applies to synthetic actors; human participants cannot be assumed to forget previous conditions.

Each assisted condition is also compared with A. These are proposed analytical contrasts, not estimates computed from human data. Task outcomes have different units and will not be pooled by subtracting raw MAB and TBS scores.

3.2 Imported cohort: 16 for development, 46 for evaluation

The importer now joins all 62 individuals across the supplied trial histories, fitted MAB parameters, reference measures and original core-profile classifications [7]. It preserves 12,400 MAB and 11,904 TBS trial rows, 200 participant-level reference measures and 56 trial-calculation definitions. The supplied development subset contains four individuals from each profile. Its deterministic selection favours clear profile membership and diversity within retained candidates, using profile scores and behavioural descriptors rather than directly ranking model fit or task performance. This clarity preference limits its representativeness. The evaluation complement contains the remaining 46 individuals. Each actor retains an individual policy for each task; a profile label is neither a generative model nor evidence of AI responsiveness.

The fitting procedure, feature definitions, penalties, source-file hashes, relevant implementation fingerprints and runtime versions are recorded before the new prediction reports are computed. Development and evaluation results are reported separately. The original profiles, development selection and original MAB model selection already used full-cohort information; the 46-person complement is therefore not independent external validation of those earlier choices. Participant separation also does not replace chronological training/test splits within an individual.

Missingness is retained explicitly: 82 MAB timeouts, 171 TBS timeouts and 138 TBS choices unavailable because of a documented logging bug. The latter occur in two evaluation-complement participants and are not recoded as timeouts or safe choices. Ineligible choices are excluded from choice likelihoods while elapsed opportunities, observed feedback and score progression remain available. No individual is excluded merely for a boundary fit or difficulty reproducing behaviour.

3.3 Two validation problems before assistance effects

First, in *next-choice prediction*, the model predicts a distribution over a participant's next recorded choice from permitted current state and earlier feedback. MAB training uses trials 1–150 and testing uses trials 151–200; TBS training uses blocks 1–16 and testing uses blocks 17–24. Fitting and standardisation use training observations only. Sequential test predictions may condition on earlier observed test feedback, never on future outcomes. Since historical TBS offers are unavailable, its predictions are marginal over unrecorded offers, not predictions conditional on the exact alternatives seen by the human.

The implemented reports use log loss and Brier score, alongside uniform choice and training-only choice frequencies with Dirichlet(1) smoothing. Cohort summaries average individual scores equally and report the two participant subsets separately. Binary TBS Brier scores and six-option MAB Brier scores use different definitions and are not pooled. Full-data reconstruction fits are exported separately from training-only validation fits; a fit on all trials is never substituted into a held-out prediction report.

Second, in *independent simulation*, each actor generates choices and updates from its own feedback under fresh outcome realisations. Full 200/192-trial runs compare performance and selected switching, exploration, feedback-adjustment and pressure/horizon measures with observed references. Parameter/policy recovery simulates from a known generating model and refits it. Recovery and human behavioural validation are complementary [11]; neither numerical convergence nor agreement on final score establishes fidelity. There is no automatic validation-success threshold in the current reports.

Profile-defining quantities remain distinguishable from validation outcomes. Reproducing a profile ingredient is not external validation of that profile. The supplied original MAB code and fits support numerical reproduction; the missing TBS offers require the explicitly provisional model in Section 4. The first results, including discrepancies, are reported in Section 6.

3.4 Matched runs, response assumptions and interpretation

In each synthetic comparison, fixed characteristics persist but experience and the task environment start afresh. Calibration observations are generated or provided separately; the assistant's calibration does not initialise the actor with experience from a previous comparison. A condition-independent keyed random source coordinates choices, environmental changes, option pairs and outcomes. Under the null scenario, where all assistance-induced response and cost parameters are zero, all conditions must yield identical choices and outcomes for the same individual, task and repetition; individual baseline omissions remain possible. When choices diverge, matching the random source does not require identical realised rewards.

Response scenarios currently include null, responsive and interference settings. Both styles use explicit parameters for responsiveness, reactance, distraction and omission. Direct advice supplies an action target; Socratic intervention invokes a task-specific reconsideration mechanism without supplying a recommended option. Reconsideration can retain the original preference. These are assumptions for sensitivity analysis, not response estimates derived from the original unassisted experiment. Offer responsiveness can be varied separately. The implemented development pilot additionally varies the reconsideration rule while holding response coefficients fixed, as detailed in Section 3.5. Actors otherwise use point estimates rather than a posterior over task parameters. Parameter uncertainty and richer dynamic responses remain further work. No single assumed mechanism is reported as a human treatment effect.

The proposed primary task outcomes are MAB total reward and TBS block target attainment, supplemented by behavioural adaptation and interference measures. Weaker-task status must be

defined from prespecified baseline evidence rather than assisted outcomes or incomparable raw task scores. Dual-regime and Low-expression profiles need not have a unique weaker environment. Repeated unassisted baselines and appropriate comparisons are needed to address regression to the mean. Replications reduce Monte Carlo uncertainty; they do not create additional independent participants. Cohort-level inference, uncertainty intervals and multiplicity control remain part of the prospective analysis plan.

3.5 Why vary Socratic reconsideration during development?

The unassisted histories constrain task behaviour but contain no observations of how the original individuals respond to advice or questions. A synthetic comparison therefore needs an explicit account of how an intervention changes a choice distribution. If Socratic assistance is represented by one favourable reasoning heuristic, an apparent advantage may largely express that authored heuristic. Choosing the heuristic after inspecting assistance outcomes would compound this problem. We consequently introduce a small, retained set of processing mechanisms during the 16-person development stage, before expanding the assistance experiment, rather than making reconsideration an unquestioned component of the full study [9]. This applies productive ignorance to the experiment itself: the unknown human response is retained as a limit on interpretation, and alternative processing rules make the model's consequences open to examination.

3.5.1 Equal response strength does not imply equal choices

We separate the strength of influence from the operation performed in response. Let p_t be the actor's baseline distribution over task options conditional on responding, and let q_t^D and $q_t^{S,m}$ be the response targets for direct advice and Socratic mechanism m. With distraction and additional omission set to zero and no directional reactance, the model uses

$$p_t^D = (1-\rho)p_t + \rho q_t^D, \qquad q_t^D = \mathbf{e}_{a_{\text{rec}}}, \tag{4}$$

$$p_t^{S,m} = (1-\rho)p_t + \rho q_t^{S,m}, \qquad q_t^{S,m} = \mathcal{R}_m(h_t, s_t, i_t). \tag{5}$$

Here $\mathbf{e}_{a_{\text{rec}}}$ concentrates probability on the recommended option; $\mathcal{R}_m$ reconsiders the issue i_t using the actor's own task parameters, permitted history h_t and public state s_t. The common coefficient ρ controls the mixture weight, not the resulting action. For a hypothetical risky-choice baseline of 0.40 and $\rho = 0.5$, advice recommending RISKY yields 0.70, whereas a reconsideration target assigning RISKY probability 0.70 yields 0.55. Identical response targets would instead yield identical updated distributions under these controls. This numerical example explains the model and is not a participant result.

Equal coefficients provide a control on configured response strength; they do not establish psychological equivalence between styles or make the mechanisms neutral. Direct advice and a question can induce different targets, be eligible on different trials and lead to different later histories. The pilot therefore records intervention counts and the resulting distributions alongside outcomes. Synthetic actors process structured issues rather than natural-language meaning, so observed style differences remain conditional on these processing assumptions.

3.5.2 Three mechanisms retained in the pilot

All three mechanisms are compared for every development individual; none is selected because it produces a desirable assistance effect.

1. **No strategic revision (`none`).** The response target retains p_t. Ordinary task learning and feedback updates continue, while the processor produces no directional revision or reactance.

Distraction and omission costs can still apply when enabled. With those additional costs zero and matched streams, D/F must reproduce A, even if their policies deliver questions.

2. **Existing public-evidence heuristic (`current`).** In MAB, exploration questions favour less-sampled arms; other issues use the existing observed-outcome value heuristic with the actor's choice sensitivity. In TBS, the heuristic considers the displayed pair, target gap and remaining horizon, using final-choice attainment when applicable. This is an authored bounded heuristic, not an optimal planning solver.

3. **Recent-feedback processing (`recent`).** In MAB, the same issue-specific heuristic uses only the last six elapsed trials. Misses consume memory positions but do not count as reward evidence. In TBS, shortening history alone would not change the current rule, which uses the present display. The alternative therefore models a perception bias: recent same-option feedback changes perceived chances before the existing goal/horizon heuristic is applied.

For TBS option a, the recent-feedback rule is

$$\widetilde{\pi}_{t,a} = \frac{4\pi_{t,a} + w_{t,a}}{4 + n_{t,a}}, \tag{6}$$

where $\pi_{t,a}$ is the displayed win chance, $n_{t,a}$ is the number of known responses to that SAFE/RISKY identity in the last six elapsed trials of the current block, and $w_{t,a}$ counts its positive gains. The rule resets at block boundaries and excludes unknown choices and misses from these counts. It transfers feedback across changing offers of the same identity. This is an explicit behavioural assumption, not a statistically warranted update of a displayed probability. The six-trial window and pseudo-count of four are fixed pilot choices, not fitted human parameters. Actual offered probabilities, reward draws, scoring and task transitions are unchanged. Recent-feedback processing may help, harm or leave choices unchanged; it is not designated a superior reasoning strategy.

These mechanisms use public displays and previously observed outcomes, without hidden MAB probabilities, future feedback or evaluator benchmarks. They are separate from original multilevel profile labels and from the assistant's uncertain participant model. Direct advice never invokes a Socratic processor. Later work can introduce learning from repeated questions, habituation, changing engagement or comprehension, informed by human interaction data. Introducing those richer mechanisms now would add unidentified assumptions without resolving the initial dependence on a processing rule.

3.5.3 Frozen pilot, matched controls and analysis

The development pilot retains the supplied 16 individuals, four per original core profile, and the full 200-trial MAB and 192-trial TBS protocols. Each individual completes all five conditions under each mechanism, initially for one repetition. Responsiveness is fixed at 0.5; reactance, distraction and additional AI-induced omission are zero, identically across individuals, styles and mechanisms. Individual task parameters and estimated baseline omission rates remain fixed at their imported values. These response settings are controlled assumptions, not estimates from neural measures, profile membership or unassisted feedback sensitivity.

The configuration defines 48 jobs (individual $\times$ mechanism $\times$ repetition), each containing ten comparison sessions and two complete unassisted calibrations: 480 comparison sessions and 96 stored calibration sessions. Calibrations are regenerated using common streams, so their numerical histories match across mechanisms for a given individual and repetition. Subsequent repetitions would draw fresh calibration and comparison histories. Actor experience resets between conditions. Mechanism names, condition identifiers and job identifiers do not enter the random streams; recorded duplicates are not additional participants.

The pilot checks three invariants: A/C/E trajectories and probabilities must match across mechanisms; under the null response setting all five conditions must match; and with zero extra costs the no-revision D/F variants must match A. These controls check the intended isolation of the processing mechanism. The no-revision variant is a negative control, not a requirement that inquiry improve outcomes when the actor makes no strategic response. Sensitivity between the two revision-capable rules is interpreted alongside that reference. Responses can subsequently change feedback and later policy decisions in D/F, so their intervention counts need not match across mechanisms.

Primary effects are paired changes from A in MAB reward and TBS target-attainment fraction, the latter calculated as attained blocks divided by 24. Reports retain individual effects, original-profile summaries, participants worse than A, intervention counts and omissions. Within each individual, effects are compared across mechanisms for ranges and sign reversals. Repetitions are averaged within people before cohort summaries. With one repetition, per-person Monte Carlo standard errors are unavailable; later repetitions address simulation variability, not uncertainty about human response or the chosen model. The analysis asks which findings persist across processing assumptions and exposes those that do not. It does not select a winning mechanism or establish a superior assistance style.

A resumable runner freezes the study configuration, actor exports, response settings, mechanism definitions, source fingerprints and software versions before execution. It verifies completed reports, preserves interrupted attempts separately and excludes partial protocols from outcome summaries. Processing targets and mechanism parameters remain in researcher-side reports rather than assistant evidence. This implementation supports the pilot without refitting the original task-policy reconstructions or changing their historical validation results [9]. The full-protocol batch was launched on 6 November 2026; completed integration checks and pending pilot outcomes are distinguished in Section 6.4.

4 Empirical actors and modelling assumptions

The empirical pipeline makes two distinct products from the supplied histories: full-data task-policy reconstructions for conditional simulation, and training-only models for prediction checks (Figure 3). Both remain researcher-side. They are not the assistant's participant estimates and are not intended as complete models of the source individuals.

4.1 MAB: reproduce the original dual-rate learning model

The imported MAB actor follows the original retained dual-rate Rescorla–Wagner model [7]. Values start at zero and evolve only for the chosen arm on an eligible, completed response:

$$\delta_t = r_t/100 - Q_t(a_t), \tag{7}$$

$$Q_{t+1}(a_t) = Q_t(a_t) + \alpha_{\pm}\delta_t, \qquad \alpha_{\pm} = \begin{cases} \alpha_+, & \delta_t \geq 0, \\ \alpha_-, & \delta_t < 0, \end{cases} \tag{8}$$

$$P(a_t = a \mid Q_t) = \frac{\exp(\beta Q_t(a))}{\sum_{b=1}^{6} \exp(\beta Q_t(b))}. \tag{9}$$

Unchosen values are unchanged. Values continue across task stages; a timeout incurs its task penalty without a learning update. The points/100 convention is explicit and is not replaced by the older illustrative actor's points/200 convention. Full-data reconstructions use the supplied fitted parameters. Distinct training-only estimates use bounded L-BFGS-B with analytic likelihood gradients, seven fixed anchor starts and 24 seeded random starts; bounds are $[.001, .999]$ for

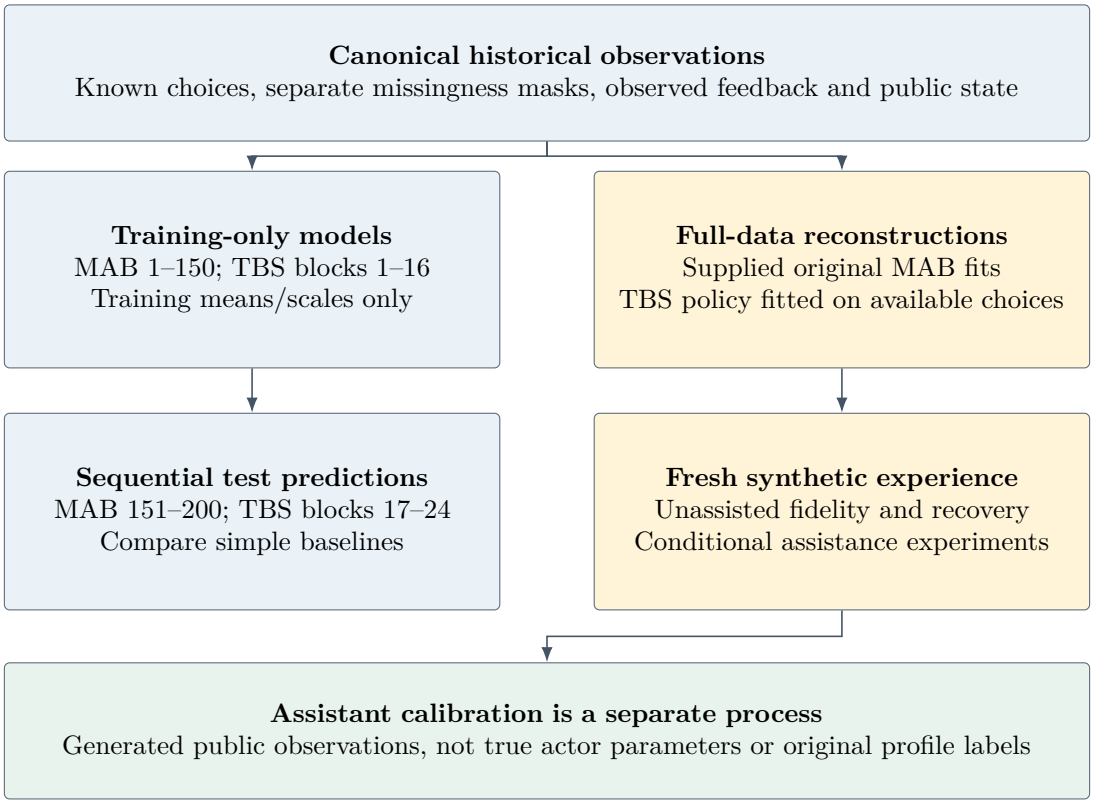


Figure 3: Separate roles for reconstruction, held-out prediction and assistant calibration. Full-data fits do not enter test prediction. The assistant receives only permitted calibration observations and public events; original profile metadata, actor coefficients and evaluator diagnostics remain outside its inputs. Historical TBS offers are unknown on both fitting branches.

each learning rate and $[.01, 30]$ for inverse temperature. The 21 supplied boundary-fit flags are retained. Neither a converged optimiser nor a boundary point identifies a psychological mechanism uniquely.

4.2 TBS: an offer-marginal state and feedback policy

The supplied materials do not unambiguously identify the historical offered pair or its per-trial probabilities and gains [7]. We leave those fields unknown. We neither reconstruct offers from received gains nor generate replacement offers for historical prediction. The source's descriptive one-predictor slopes are also not added together as if they formed a jointly fitted choice policy.

Instead, the provisional TBS actor estimates risky choice from current state and immediately preceding same-block feedback. Let T_t be target, S_t accumulated raw points, h_t remaining opportunities including the current choice, m_t the final-win multiplier, $g_t = \max(0, T_t - S_t)/h_t$ and $f_t = \mathbf{1}[h_t = 1]$. The seven predictors are

$$x_t = (g_t, h_t, f_t, f_t(T_t/m_t - S_t), b^r_{t-1}, b^k_{t-1}, b^+_{t-1}). \tag{10}$$

The three final indicators describe a previous risky choice, a previous known choice and a previous positive gain. They reset at block boundaries. The known-choice indicator distinguishes a known safe response from an unavailable preceding choice; independently observed feedback can remain usable when that choice is unknown. The final-required-gain feature describes the gain needed if a final win activates the multiplier; it is zero before the final opportunity.

For standardised features $\tilde{x}_t$, risky choice has probability $p_0(s_t) = \sigma(w_0 + w^\top \tilde{x}_t)$, where σ is the logistic function. The fit minimises negative log likelihood plus

$$\tfrac{1}{2}\left(0.001 w_0^2 + \sum_{j=1}^{7} w_j^2\right). \tag{11}$$

Means and sample standard deviations come only from eligible fitting observations; constant predictors use scale one. These regularisation and feature choices are authored modelling assumptions. Full-data and training-only policies are fitted separately. This is an offer-marginal policy, not an established original TBS cognitive model or evidence that people ignore the alternatives.

Prospective offer sensitivity is varied without silently changing that fitted marginal. For fresh pair j, define $d_j = (\mathrm{EV}_{r,j} - \mathrm{EV}_{s,j})/100$. The conditional policy is

$$p_\eta(s, j) = \sigma(c_\eta(s) + \eta d_j), \tag{12}$$

$$\frac{1}{8}\sum_{j=1}^{8} p_\eta(s, j) = p_0(s). \tag{13}$$

The offset $c_\eta(s)$ is solved numerically for each state. Default $\eta = 0$ applies the fitted marginal; $\eta = 2$ is an illustrative sensitivity setting, not an estimated human coefficient. The contrast concerns immediate expected gain, not optimal multi-step target planning. Fresh offers and target/multiplier combinations are drawn independently and uniformly from the current generator's eight pairs and four combinations. Original realised offer schedules and hidden MAB changes are not replayed.

4.3 Omissions, calibration and provenance

For each person and task, baseline omission uses the posterior mean $(u + 0.5)/(n + 1)$ under a $\mathrm{Beta}(0.5, 0.5)$ prior, where u is recorded timeouts and n elapsed trials. Full-history rates

belong to reconstruction actors and training-only rates to validation actors. Future omissions are independent Bernoulli draws. Baseline and additional AI-induced omissions combine as $1 - (1 - q_{\text{base}})(1 - q_{\text{AI}})$. These are explicit assumptions, not a fitted model of reading, reaction time or workload.

The assistant fits its own existing coarse models from separately generated unassisted calibrations. It receives neither true actor coefficients nor original profile labels/core scores, neural or dispositional measures, future-linked diagnostics or ideal-observer benchmarks. Its calibration can therefore misrepresent the actor's richer policy. Poor personalisation under this mismatch would not establish that personalisation or Socratic inquiry cannot help humans.

Each generated bundle records the fitting procedure, missingness rules, source and implementation fingerprints, software versions and a machine-readable assumptions register [8]. Frozen artefacts are checked for drift before export and analysis. Point estimates, offer sensitivity, AI responsiveness, timing and inference limits have separate entries, so data-constrained quantities remain distinguishable from assumed mechanisms. Researcher-side histories and parameters remain local rather than entering the public assistance trace.

5 Architecture: separate epistemic and behavioural responsibilities

The preceding comparisons require independent control of how a participant acts, what the assistant can observe, which interpretations the evidence supports and whether communicating an interpretation is worthwhile. If these responsibilities are conflated, an apparent assistance benefit could arise from access to hidden task information or an assumed participant response, while a fluent explanation could obscure the uncertainty behind an intervention. The architecture therefore gives productive ignorance a concrete role in the experiment: limitations in evidence constrain the reasons available to the assistant, unresolved objections constrain what it may claim, and interruption costs constrain whether it should speak. Separating these responsibilities also allows participant models, reasoning rules and delivery policies to be refined and evaluated independently.

Figure 4 shows how these separations organise the interaction loop. The upper, gold-coloured components belong to the researcher-side simulation. The private environment generates task states and outcomes; the synthetic participant makes choices using its own task policy, experience and assumed response to assistance; and the evaluator retains the private information needed to interpret performance and paired comparisons. The validated public-event interface below them is the boundary between this machinery and the assistant. It admits permitted task displays, observed choices and outcomes, and attributed participant reports, while withholding hidden reward probabilities, future outcomes and the actor's latent parameters. The condition filter additionally controls the use of participant estimates, so that personalisation in E/F depends on uncertain calibration rather than privileged knowledge of the simulated individual.

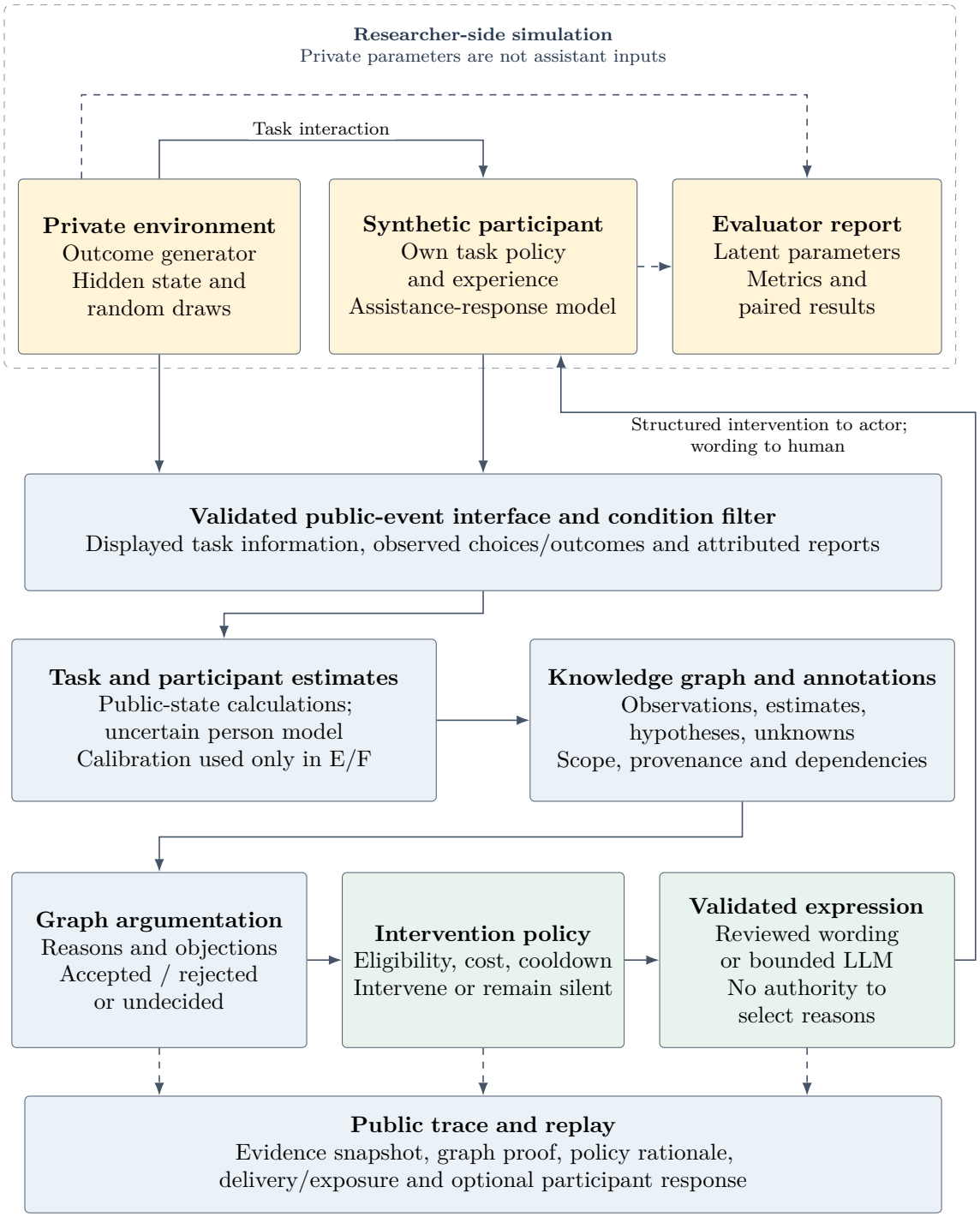


Figure 4: Architecture separating participant behaviour, epistemic assessment and intervention delivery. Gold boxes contain private simulation components; the public-event interface restricts the evidence available to the assistant. Below that boundary, numerical models inform an annotated knowledge graph, argumentation evaluates reasons and objections, and a separate policy determines whether and how to intervene. Validated expression returns the intervention to the participant, while the trace retains its evidential and delivery history. Solid arrows show operational flow; dashed arrows show evaluator or trace records.

Below the information boundary, task calculations and uncertain participant estimates populate the knowledge graph with their provenance, scope and epistemic status. Graph argumentation then constructs supporting reasons and objections and evaluates arguments as accepted, rejected or undecided. This is the epistemic part of the architecture: it determines which grounds for

communication survive the current evidence and rules, without turning an accepted model-based inference into an observed fact. The intervention policy performs a different assessment, combining condition eligibility, relevance, interruption cost and recent delivery history. An accepted reason may consequently produce no message, while a permitted unresolved conflict may support a clarification question in D/F. The final expression component renders the selected intervention through reviewed wording or a bounded language-model mode; it has no authority to invent reasons or settle objections.

The return arrow closes the behavioural loop: a synthetic actor receives a structured intervention processed by the separate assistance-response model, whereas a human participant encounters its wording in the task interface. Subsequent choices and optional responses supply new public observations for later assessments. Along the bottom of the figure, the trace records the evidence snapshot, argument evaluation, policy rationale and actual delivery or silence. This makes it possible to distinguish an unsupported interpretation from an unwarranted interruption or a failure to expose the selected message, rather than attributing every outcome to the advice or question alone. In human sessions, the person replaces the synthetic actor while the assistant retains the public-event interface and reasoning pipeline; the simulated response mechanism does not become a model of how that person understands the message.

5.1 Actor, assistance response and assistant model

The synthetic actor has its own task parameters and history. Environment-owned adapters now support the imported original MAB model and the provisional TBS state/previous-feedback policy described in Section 4, alongside the earlier illustrative actors. Task-specific empirical import and fitting sit outside the shared runner; additional environments can provide their own policies without adding task branches to the shared engine or intervention policy. Original multilevel profiles remain researcher metadata, rather than rules that determine assistance responsiveness.

The assistance-response model modifies the actor's choice distribution only when an intervention is delivered. It is distinct from the assistant's participant model. The latter contains estimates from unassisted calibration and permitted observations, not the actor's true learning rates or responsiveness parameters. Giving those latent parameters to personalised assistance would confer an unrealistic information advantage.

Socratic processing is a separate researcher-controlled component within that response model. Its selector supports no revision, the current task heuristic and an optional task-owned recent-feedback rule. An unsupported variant fails before sessions are created; adding an environment-specific rule does not require task branches in the shared actor or intervention policy. The selector and its diagnostics remain outside the assistant's information boundary. Section 3.5 describes why varying this component complements, rather than replaces, sensitivity to response strength and cost.

Interaction estimates are intentionally narrow. Explicit reports can describe burden and can adjust interruption cost in E/F. Agreement with a delivered direct recommendation is recorded as concordance, not measured trust, causal compliance or benefit. Socratic reconsideration remains unknown without a report. A disagreement can contest an interpretation without establishing a general aversion to assistance. Personalisation influences intervention selection through participant evidence, without presuming that personalised support is more useful.

5.2 Knowledge representation and graph reasoning

The knowledge representation connects a proposed intervention to the task state, observations, calculations and assumptions on which it depends. Three complementary elements support this connection:

- **Persistent records.** Canonical JSON events and snapshots are persisted locally in SQLite, including numerical model outputs and delivery-policy records.
- **Graph projection.** A deterministic projection exposes selected records as an RDF graph of subject–predicate–object triples with identifiable resources and typed values [13].
- **Shared and task-specific vocabularies.** RDF/OWL vocabularies organise a shared epistemic core, a task-independent layer and separate `mab:` and `tbs:` namespaces; an `arg:` extension represents propositions, arguments and attacks.

The vocabularies specify the kinds of knowledge the prototype can represent. They do not constitute an unrestricted OWL inference engine: numerical inference runs in Python, while the graph reasoner executes explicitly authored argument schemes.

5.2.1 Represented knowledge and epistemic status

Table 2 shows how each category of knowledge is encoded and linked to its context and provenance:

Knowledge	Graph representation	Context and provenance
Mechanics and state	`task:TaskDefinition`, `task:Rule`, `task:Option` and `task:TaskState`; domain properties describe arms, gambles, points, targets and horizon.	Task version, trial, phase and previous state.
Observations	`task:ChoiceObservation` and `task:OutcomeObservation`; selected option, reward, timeout and response time.	Source trial and the choice to which an outcome belongs.
Calculations and estimates	`task:Calculation` and `task:Estimate`; typed values, units, sample counts and available uncertainty bounds.	Input state, named versioned rule, option and supporting observations.
Claims and reports	Core claims and hypotheses, plus `arg:Assertion` records with scoped propositions and attributed participant reports.	Source, epistemic status, dependencies, contestation and revision links where applicable.
Arguments and attacks	`arg:Argument` and `arg:Attack`; premises, conclusion, subarguments, scheme, attack target and evaluation label.	Rule identity, inference kind, priority and links to the intervention or considered objection.

Table 2: Knowledge represented in the assistant-side graph. Numerical models compute values before projection; the graph preserves their meaning and provenance. Candidate scores, interruption costs and cooldowns remain in the canonical JSON policy records.

- **Mechanics and state.** Public task definitions and current decision contexts include MAB arms, displayed TBS gambles, targets and conditional scoring rules. Dynamic states identify the trial and phase and describe quantities such as accumulated points, target gap and remaining choices.
- **Observations.** Records of selected options, received rewards, timeouts and response times

describe what occurred in the task. Choices and outcomes are represented separately and remain distinguishable from calculations or interpretations based on them.

- **Calculations and estimates.** Derived quantities include MAB sampled-arm counts, reward beliefs, change estimates and learned values, as well as TBS expected gains and final-choice attainment probabilities. Estimates of the participant's task policy also belong here and retain their evidence and uncertainty.
- **Claims and reports.** Claims express task interpretations and provisional behavioural hypotheses based on the available evidence. Attributed participant reports can supply a stated strategy or contest an interpretation; their source and epistemic status distinguish them from observations and model outputs.
- **Arguments and attacks.** Explicit reasoning records connect premises to conclusions through named schemes and retain objections to premises, conclusions or rule applicability. Their evaluation labels show which grounds for communication are accepted, rejected or unresolved.

Hidden MAB probabilities, realised change times and future outcomes remain outside this assistant-side representation.

A derived quantity links to its input state through `task:inputState`, to the calculation rule through `task:usesRule`, and, where relevant, to an option or supporting observation. Thus a reward estimate can be traced to sampled outcomes, while displayed-input arithmetic can be traced to the current option and published scoring rule. Claims carry epistemic annotations such as known, unknown, assumed, inferred, contested, probable, low-confidence and source-dependent; these need not be mutually exclusive. Relations including `supportedBy`, `dependsOn`, `contradicts` and `revises` retain evidential and revision history. Unknown is not interpreted as false, and a participant's disagreement is recorded as an attributed report rather than silently replacing an observation or numerical posterior.

For reasoning, an `arg:Assertion` wraps a structured proposition $\langle s, p, o, \sigma, \epsilon \rangle$: subject, predicate, typed object, current-state scope and positive or negative polarity. The assertion separately records its kind, source and epistemic statuses. For example, a TBS state's `tbs:points` value can be a known observation, whereas a pressure-sensitivity estimate is inferred and source-dependent. Accepting an argument that uses the latter does not establish an observed disposition. Explicit scope also prevents a comparison made before a choice from being attributed to the state after its outcome.

5.2.2 Argument construction and evaluation

The implementation separates model calculation, argument discovery and argument evaluation. Python computes Bayesian reward estimates and task arithmetic; an optional pgmpy backend supports a small interpretation network with authored probability tables. These numerical outputs supply premises, not argument priorities or diagnoses. The graph pipeline then proceeds as follows:

1. **Project the current evidence.** The reasoner builds a local graph from the current task snapshot, permitted history aggregates, eligible model hypotheses and scoped participant reports. Numeric state, option, goal and calculation records become explicit assertions. Re-analysis of an earlier event uses only its recorded history prefix, excluding later feedback or reviews.
2. **Instantiate authored schemes.** Locally versioned SPARQL `SELECT` queries [14] run through RDFLib to find premises satisfying a scheme's graph patterns and numerical

conditions. Each match must return existing assertion identifiers from the same state. It creates an argument recording its premises, conclusion, subarguments, rule identity and strict or defeasible inference kind. Derived assertions can support further schemes in bounded forward rounds; circular self-support is excluded. Queries cannot execute updates or remote services, and participant or language-model text does not become an executable rule.

3. **Construct attacks and defeats.** An argument can undermine a contrary estimated, reported or assumed premise, rebut a contrary defeasible conclusion, or undercut the applicability of a named defeasible scheme. Contrariness requires explicit opposite polarities over the same scoped proposition; missing evidence and different decision criteria do not count as contradictions. Successful attacks propagate to arguments containing the defeated subargument. Authored priorities govern rebuttal and undermining; undercutting is independent of those priorities. The initial schemes use equal priorities, independently of Bayesian confidence and intervention usefulness.

4. **Compute grounded labels.** A finite fixed-point procedure accepts an argument when all its defeaters are rejected, rejects it when a defeater is accepted, and leaves unresolved dependencies or cycles undecided [15]. The epistemic status known is assigned only to strict conclusions with exclusively known premises; other derived conclusions remain inferred. Argument labels express justification under the recorded premises and rules, rather than truth or measured usefulness. The subsequent policy can select only condition-permitted, proof-backed communication opportunities.

Python validates schemas, arithmetic, proof links, acyclicity and state consistency. When pySHACL is installed, SHACL additionally checks constraints on the RDF projection [12]; this checks structural conformance rather than empirical truth. Current discovery limits are six rounds, 256 arguments, 512 assertions and 1,024 query matches per snapshot. Reaching a construction limit marks the analysis incomplete and all argument labels undecided, because an undiscovered objection might change an apparent acceptance; the delivery policy then remains silent. Unavailable RDF reasoning likewise produces no intervention. Fixed rule order, sorted matches, content-derived identifiers and frozen configuration hashes support replay. These bounds and checks constrain execution without establishing real-time performance or scientific validity.

5.2.3 Two illustrative reasoning traces

The implemented example runner generated these synthetic traces in unassisted calibration, where condition A remains silent. Their possible use in assisted conditions is subject to the separate delivery policy.

MAB: limited coverage does not settle the value of exploration. After seven observed choices have sampled only one of six arms, a strict coverage scheme derives that alternatives remain unsampled. A defeasible scheme then concludes that further exploration is worth discussing; it does not conclude that the participant has made an error. In a later snapshot with only two choices remaining, a second defeasible scheme uses the short horizon to oppose that same conclusion. With equal priorities, both exploration arguments remain undecided, while the arithmetic coverage conclusion remains accepted. A separate short-horizon undercutter blocks a personalised challenge based on underexploration. In D/F, an explicitly authored clarification may nevertheless ask, "With so few decisions remaining, what would you gain from trying another option?" C/E cannot derive advice from the undecided exploration warrant, although unrelated accepted reasons remain available. Productive ignorance here preserves both the information gap and the objection to acting on it.

TBS: compatible calculations can favour different choices. Before the final choice, suppose the accumulated total is 200 points, the target is 930 and the multiplier is 2.3. The displayed alternatives offer a gain of 180 with probability 0.60 and a gain of 260 with probability 0.35. Python computes immediate expected gains and totals conditional on a final win:

$$\mathrm{EV}_{\mathrm{safe}} = 0.60 \times 180 = 108, \qquad \mathrm{EV}_{\mathrm{risky}} = 0.35 \times 260 = 91,$$
$$W_{\mathrm{safe}} = (200 + 180) \times 2.3 = 874, \qquad W_{\mathrm{risky}} = (200 + 260) \times 2.3 = 1058.$$

The final-choice attainment probabilities are therefore zero and 0.35, respectively: a loss leaves 200 points and does not activate the multiplier. The graph stores expected-gain and attainment calculations as distinct typed resources, each linked to the input state, option and rule, with units of points or probability. Strict schemes establish the two rankings and then derive `tbs:criteriaDiverge`. These conclusions are compatible because they concern different criteria. A further defeasible scheme supports clarifying the objective, without claiming to know the participant's intention. Its reviewed Socratic wording asks, "Expected gain and final target attainment favour different options. Which objective guides your choice?" C/E require an additional accepted option-comparison proof to recommend an action under their stated criterion. The trace thus exposes how a mathematically correct calculation can support a question about the grounds for choosing, rather than manufacture a single unconditional answer.

5.3 From justified reasons to justified interruption

Graph argumentation is the single source of live reasons and objections. The policy selects among graph-grounded communication opportunities; it cannot invent a separate reason when graph reasoning fails. Supported content and a useful interruption remain different questions (Figure 5). Usefulness, interruption cost, thresholds and cooldowns are explicit, provisional policy parameters. The policy does not establish the acceptance of an argument.

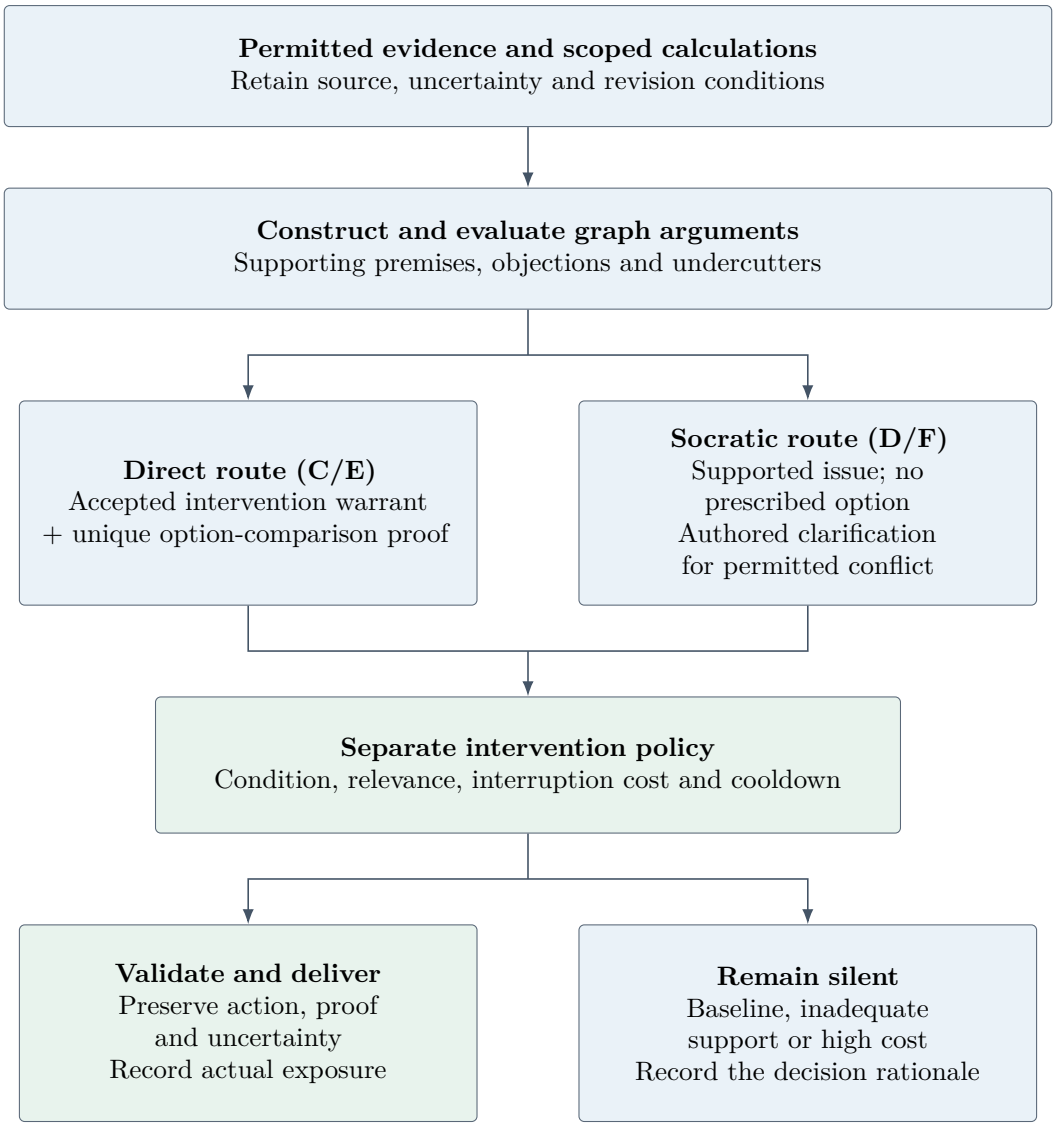


Figure 5: Acceptance of an argument does not compel an intervention. Direct advice requires a separate comparison proof. Socratic clarification of unresolved conflict is permitted only by an explicit authored scheme. Incomplete or unavailable reasoning produces no live intervention candidate.

Direct assistance identifies an available option, response key and criterion. MAB compares posterior mean rewards estimated from observed outcomes; this is not an optimal exploration policy. TBS compares immediate expected gains before the final trial and exact one-step target-attainment probability on the final trial, including the conditional multiplier. A strict comparison requires a unique winner. These criteria do not solve multi-trial planning over unknown future options. Socratic interventions instead invite examination of supported issues or uncertainty. A question has no structured recommended option, although its possible indirect steering remains an empirical concern.

Reviewed templates express the selected intervention by default. An experimental local Ollama mode can compose bounded questions for D/F; C/E retain reviewed direct wording. Language packages contain the selected proof closure and actual attackers, not the whole researcher graph. Validation checks grounding and output structure before delivery. Model provider, model version and language mode must be fixed or treated as explicit factors in an empirical experiment. The model does not select graph labels, intervention policy or task actions.

5.4 Timing, exposure and participant contestation

The browser interface currently inserts a four-second preparation window in all conditions, including A, followed by the five-second choice window. Delivered messages remain visible through the trial and are dimmed as previous messages afterwards. Optional reply and feedback controls remain visible but accept new responses only for an active intervention in the current trial. Replies do not pause task timers and do not receive an automatic conversational answer. The trace records latency, actual exposure and early choice attempts.

The current TBS backend receives the upcoming option pair during preparation, while the participant sees progress and target information until choices open. This timing is not equivalent to simultaneous information availability. Before a human experiment, exposure must be reconciled so that an intervention does not require a participant to assess an alternative not yet visible. Labels describing intervention eligibility must also match actual screen exposure. Equal nominal windows do not guarantee equal reading opportunities; realised exposure and delivered intervention counts must be reported.

5.5 Extensibility and reproducibility

Each registered environment owns event validation, mechanics, numerical models, ontology documents, argument schemes, recommendation wording, actor policy, private simulator, metric definitions and browser adapter. A study JSON selects environments, per-environment protocols, calibration dependencies and conditions. Adding an environment requires its module and assets, registration and study configuration rather than new branches in the shared engine or policy. Metrics declare units and whether to maximise or minimise a quantity.

A third, four-trial REFERENCE environment exercises this contract with its own namespace and an error metric to minimise. It is an engineering fixture, not an additional empirical decision task. Implemented checks cover all five conditions, actor resets, matched null-response behaviour, public/private separation, graph provenance, task completion and replay. Scripted interface checks use actual task scripts with Python persistence; they are not visual or laboratory timing validation. No human assistance effect follows from passing these checks.

Session configurations freeze task, ontology, graph, model and policy descriptors as applicable. Separate evaluator reports retain private specifications, response assumptions, realised choice probabilities, metrics and paired differences. Execution-adapter source fingerprints support drift detection; reproducing a simulation still requires the corresponding code and evaluator inputs. Replay of public events and assistant decisions is distinct from reproducing latent actor choices.

6 Preliminary computational checks

The results below distinguish the empirical pipeline checks of 5 November 2026 [8] from the subsequent reconsideration integration check of 6 November 2026 [9]. They assess numerical reproduction, prediction, synthetic behaviour and integration. They are not human assistance outcomes or a confirmatory test of the factorial contrasts. Full-protocol reconsideration-pilot outcomes remain pending at this draft's cutoff.

6.1 Source reproduction and held-out choice prediction

For all 62 original MAB fits, the imported forward calculation reproduces stored pre-choice values, choice probabilities, prediction errors and likelihoods; the maximum negative-log-likelihood difference is approximately 1.14×10^{-13}. TBS raw and conditional-multiplier score arithmetic

agrees for all 11,904 supplied rows. Recomputing selected behavioural summaries produces 928 comparisons against source reference values, with maximum absolute residual 5.09×10^{-8}. This covers selected metrics, not a reimplementation of all 200 reference measures. The original MATLAB optimiser was not rerun; the check independently reproduces its forward calculations in Python.

Table 3 reports chronological prediction from training-only fits. MAB improves on uniform choice and the frequency baseline on average in both subsets. TBS improves on uniform choice but is worse than the frequency baseline in both subsets. In the evaluation complement, 28 of 46 MAB models and 18 of 46 TBS models beat their respective frequency baselines. No model changes were selected against the evaluation-46 results to erase the TBS finding.

Task	Subset	Test choices	Log loss	Frequency	Brier
MAB	Development ($n = 16$)	800	1.4904	1.5670	0.7181
MAB	Evaluation ($n = 46$)	2,295	1.6109	1.6572	0.7663
TBS	Development ($n = 16$)	1,018	0.6183	0.5850	0.2128
TBS	Evaluation ($n = 46$)	2,878	0.5806	0.5518	0.1931

Table 3: Held-out next-choice prediction. Loss and Brier columns are equal-weight means of participant scores; "Frequency" is training-only smoothed choice-frequency log loss. Lower is better. Uniform log loss is 1.7918 for MAB and 0.6931 for TBS. Timeouts and unknown choices are excluded from choice scoring and counted separately. Binary and six-option Brier definitions differ.

The TBS result limits interpretation of subsequent simulations. Missing offers prevent exact offer-conditional validation, and the provisional state/feedback model does not establish that it captures an individual's policy better than a simple marginal preference. The 46-person complement remains an internal evaluation of the frozen new procedure, with the source-selection limitations described above. Model improvement on development data and genuinely independent evaluation remain research needs.

6.2 Fresh behaviour, offer sensitivity and internal recovery

Each reconstruction was simulated unassisted for ten full-protocol repetitions per task under each of $\eta = 0$ and $\eta = 2$. This yields 2,480 recorded runs; MAB trajectories repeat as matched controls across the two settings, and repetitions are not additional participants. At $\eta = 0$, mean MAB score is 7,705.5 points versus 7,898.4 observed. Mean TBS block attainment is 0.6605 versus 0.6714 observed, and risky-choice rate is 0.2963 versus 0.2977 observed. At $\eta = 2$, the corresponding TBS rates are 0.6620 and 0.2957. These are cohort averages, not evidence of individual fidelity or successful intervention. Selected switching, exploration and feedback-adjustment discrepancies remain in the per-person reports. Matching aggregate rates does not override the negative TBS prediction result.

Internal recovery used one generated full-protocol dataset per task for each development individual, producing 32 refits. Mean policy-probability root-mean-square error on common histories is 0.0195 for MAB and 0.0868 for TBS. Because TBS scaling changes across fitted samples, probabilities rather than differently standardised coefficients are compared. This is recovery under the assumed generating models, not validation against humans, proof of identifiability or a parameter uncertainty interval.

6.3 Assistance integration and remaining experimental work

Two reconstructions completed all five conditions in both short tasks: 20 comparison sessions plus four calibrations. All 24 public sessions replay exactly. Under null responsiveness, choices

and outcomes match across conditions, including baseline omissions; public records contain no true actor coefficients or original profile labels. A non-null interference check changed 37 choice distributions across 20 comparisons capped at 12 trials; these incomplete runs were correctly excluded from paired outcome analysis. They demonstrate execution of the assumed response mechanism, not observed human interference.

The normal event pipeline also generated two labelled short unassisted calibrations for every reconstruction, giving 124 completed sessions for the known-pseudonym selector. These are generated observations, not original human trials relabelled as synthetic. Their use in a manual browser session does not automate the human's subsequent choices. Backend and scripted interface checks passed, with one socket-binding test unavailable; native browser appearance and laboratory timing were not verified in this check.

No complete full-cohort assisted experiment has yet been run, and no human response to AI is estimated from these unassisted data. The current evidence supports inspecting conditional scenarios and their failure modes. It does not establish a superior assistance condition, a causal benefit, calibrated trust or a change in neural/dispositional profile.

6.4 Reconsideration integration and pilot status

The completed reconsideration check used one development reconstruction, both short tasks, all five conditions and all three mechanisms: 30 complete comparison sessions and six complete calibrations. All six A/C/E invariance checks across mechanisms passed. The four no-revision D/F comparisons matched A under zero extra costs. Current and recent processing produced different choice distributions, while some final scores coincided; a change in a processing rule does not require a change in every realised choice or score.

All 30 public comparison sessions excluded private mechanism diagnostics and true actor parameters. Six sampled assisted sessions, one per task and mechanism, replayed exactly; earlier seeded MAB and TBS calibrations also replayed. Full-length unassisted trajectories checked for the same reconstruction still matched the original frozen baseline records. The updated backend suite passed 124 tests with one socket-binding test skipped, and the scripted participant-selector, inspector and MAB/TBS teammate checks passed. These are software and information-boundary checks, not validation of a human reconsideration mechanism.

The 48-job full-protocol development pilot is running separately under the frozen specification in Section 3.5. This manuscript reports its rationale, design and completed integration checks, with aggregate pilot outcomes pending. Partial attempts are preserved for inspection and excluded from outcome summaries. The small check is not used to choose the most favourable mechanism, estimate a style advantage or infer human responsiveness. Subsequent reporting will retain all three mechanisms, their paired effects and any sign reversals, together with the existing TBS model and timing limitations.

7 Human implications for national security and governance

The controlled tasks isolate uncertainty and changing constraints, but omit institutional authority, contested objectives, strategic adversaries and the distribution of harms. We therefore offer design questions for later professional studies, not claims of operational readiness.

Intervention authority. A justified proposition is not automatically a justified interruption. In professional settings, the policy will also need mandate, role and escalation constraints. A logged usefulness score cannot authorise a system to determine public objectives or operational priorities.

Contestability and accountability. Provenance and retained objections can make a particular intervention examinable. Optional disagreement and correction support contestation at the interface. Institutional accountability additionally requires identified decision rights, review procedures and responsibility for model and policy revisions; logs alone do not supply them.

Personalisation and agency. A participant model can expose blind spots, but can also stereotype an individual or tailor persuasive pressure. Keeping estimates uncertain, task-specific and separate from latent parameters is one constraint. Whether this preserves agency must be evaluated with people, rather than inferred from interface labels or recommendation concordance.

Preserving uncertainty under pressure. Productive ignorance does not require endless deliberation. Time-critical work may favour concise qualified advice, escalation or silence. The research question is whether the system helps a person recognise which uncertainties matter within available time, while retaining alternatives for later review.

A subsequent human study will require consent and appropriate review, data minimisation, prespecified endpoints, sample-size planning and random allocation or a justified counterbalanced design. Synthetic resets cannot remove human carryover learning. Workload, comprehension, reasoning quality and appropriate reliance will require measures beyond the current synthetic score reports. Transfer should proceed through expert-designed scenarios with domain-specific goals and evidence rules, followed by independent evaluation.

8 Current limits and next research steps

The software now provides an inspectable intervention pipeline, a matched simulation framework and individual task-policy reconstructions for all 62 source participants. Original MAB calculations are numerically reproduced; a provisional TBS policy, training-only prediction and internal recovery checks are implemented. This is a partial behavioural reconstruction, not a validated reproduction of the people, their neural profiles or their responses to AI. In particular, TBS fails to beat the frequency baseline on average despite agreement on some cohort-level behaviour. Current interpretation probabilities, argument schemes, thresholds and assistance-response mechanisms remain authored assumptions. Structural validation and replay constrain software behaviour, not scientific validity.

Synthetic response times and reading exposures are currently placeholders rather than validated models of human timing or workload. Synthetic actors process structured interventions, not natural-language meaning. Differences between assistance styles therefore depend on the response and reconsideration mechanisms. The three-variant pilot makes part of that dependence explicit while holding configured strength and costs fixed; it does not supply an empirical account of comprehension, trust or learning from questions.

Historical TBS offers cannot currently be recovered; the model proceeds with that limitation recorded rather than postponing implementation or manufacturing evidence. The immediate next evidence is the development pilot's comparison across the retained processing assumptions. Further steps are to examine individual discrepancies on the development subset, compare defensible alternative TBS policies, expand parameter and response sensitivity analyses, and secure independent evaluation where possible. The already inspected 46-person result must not be repeatedly tuned into apparent validation success. Point-estimate uncertainty, prespecified fidelity criteria, cohort-level inference and a human study of reading, comprehension and response remain unresolved. Assistance comparisons should initially be interpreted only as conditional scenarios, with model inadequacy visible alongside their outcomes. The eventual submission should add the agreed human-study protocol and uncertainty analyses, confirm internal-source references and authorship, and justify relevance beyond the controlled tasks.

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